Supplementary material

Trends in infant mortality and provincial inequalities in Argentina, 1980–2024

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Supplementary methods

Trend analysis. The procedure followed the authors' standard protocol: (1) the original series and three smoothed versions (LOESS with span 0.3 on the log scale, cubic smoothing spline and three-year centered moving average) were computed; (2) breakpoints were identified on the log-transformed original series with the Bai–Perron dynamic programming algorithm (R package strucchange), scanning from the most recent to the earliest year, with the number of breaks selected by the Bayesian information criterion and a minimum segment length h of 5 years (1980–2024) or 4 years (2005–2024); (3) within each period, the APC was estimated by ordinary least squares on the log rate, with Student t-based 95% confidence intervals; consecutive periods share the breakpoint year; (4) the same periods were analyzed with each smoothed series. Results in the main text refer to the original series, because smoothing induces autocorrelation and yields overly narrow confidence intervals.

Gini index. Provinces were ordered by decreasing rate; the Lorenz curve of the cumulative proportion of deaths against the cumulative proportion of live births was smoothed by nonparametric kernel regression (R package sm, bandwidth selected with h.select), and the area under the curve was computed with the trapezoidal rule; Gini = $|2 \times (AUC - 0.5)|$. Confidence intervals were obtained by parametric bootstrap, resampling deaths in each province from a binomial distribution with the observed rate (100 replicates for annual values and 1 000 for triennia), using the 2.5 and 97.5 percentiles.

Coefficient of variation net of noise. For each year, $CV^2observed = \Sigma\, n_i\, (r_i - \bar{r})^2 / \Sigma\, n_i / \bar{r}^2$, where r_i and n_i are the IMR and live births of province i and $\bar{r}$ the national rate. Under a Poisson model with a common rate, the expected variance of r_i is $\bar{r}/n_i$, giving $CV^2Poisson = \Sigma\, n_i\, (\bar{r}/n_i) / \Sigma\, n_i / \bar{r}^2$. The CV net of noise is $\sqrt{\max(0, CV^2observed - CV^2Poisson)}$. As births fell, CVPoisson increased from 0.050 in 2005 to 0.083 in 2024.

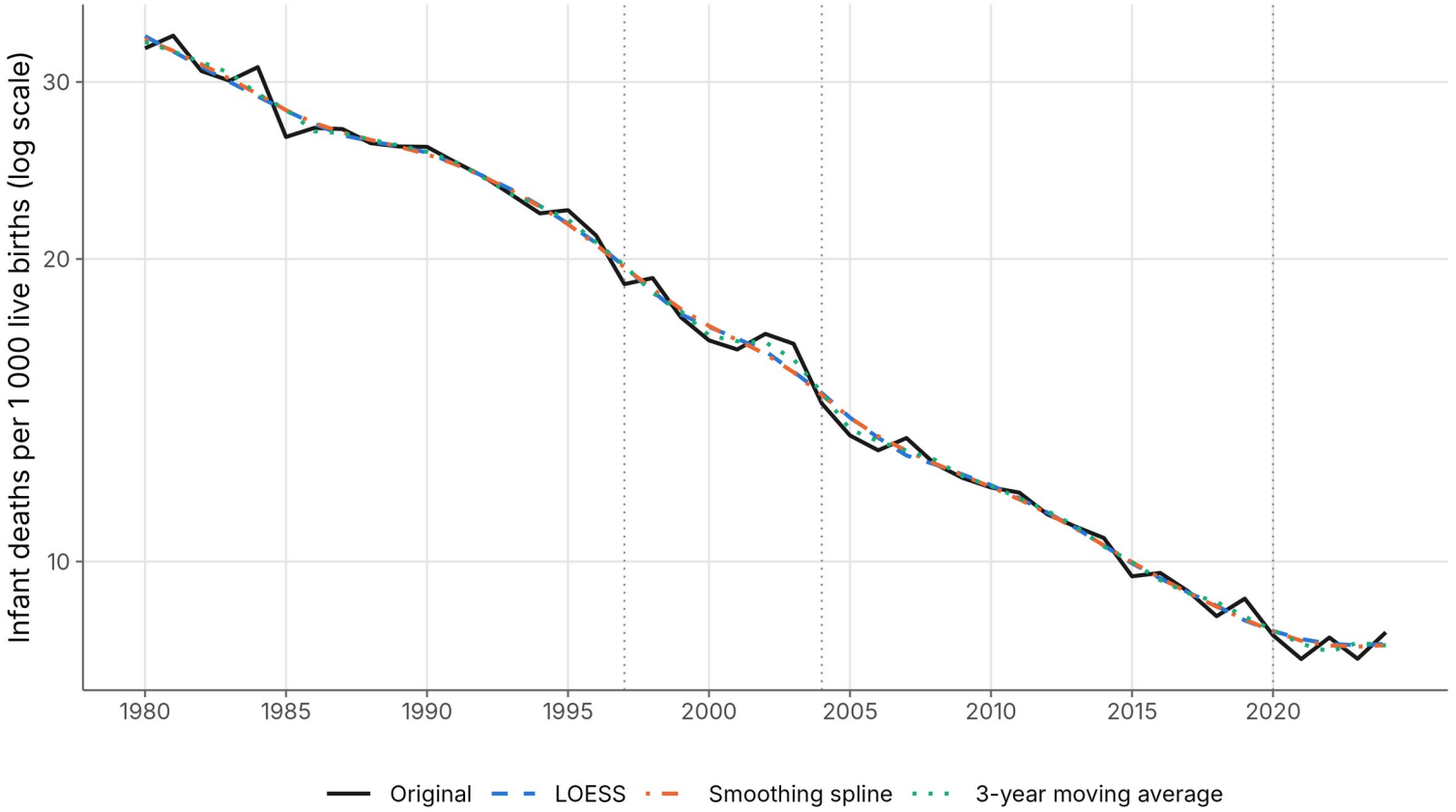
Social inequality. Provinces were ranked by the percentage of households with UBN (linear interpolation between the 2001, 2010 and 2022 censuses). The concentration index was computed with provinces ordered from highest to lowest UBN. The SII was the slope of a births-weighted linear regression of the rate on the ridit (midpoint of the cumulative share of live births) of UBN; the RII was the exponentiated slope of a quasi-Poisson regression of deaths on the ridit with log live births as offset. 95% CIs were obtained from the regression models (SII, RII) and by bootstrap (concentration index, 1 000 replicates).

Expected deaths. For each series, a quasi-Poisson model $\log E(d_t) = \log(n_t) + \beta_0 + \beta_1 t$ was fitted to 2005–2019. For 2020–2024, 10 000 vectors of coefficients were drawn from their estimated multivariate normal distribution, the expected counts were computed, and observed counts were simulated from a negative binomial distribution with variance $\varphi\mu$ (φ: estimated dispersion). The 2.5 and 97.5 percentiles of the simulated annual and cumulative counts define the 95% prediction intervals.

Birthweight data. National live births and infant deaths by birthweight for 2011–2024 were extracted from the DEIS yearbooks (Estadísticas vitales. Información básica, tables of live births by birthweight and infant deaths by birthweight). Earlier yearbooks used a different layout and were not included. The percentage of infant deaths with unspecified birthweight decreased from 20.2% in 2011 to 7.7% in 2024.

Reproducibility. All analyses were run in R 4.3.3 with scripts 01_preparar.R, 02_analisis.R, 04_analisis_rpsp.R and 05_figuras_rpsp.R (available in the study repository; see Data availability in the main text).

Figure S1. Infant mortality rate, Argentina, 1980–2024: original and smoothed series (sensitivity analysis)



Note: LOESS: locally estimated scatterplot smoothing (span 0.3, log scale). Dotted vertical lines: breakpoints identified on the original series (1997, 2004 and 2020). The same periods were analyzed with each smoothed series. Source: prepared by the authors from DEIS data.

Table S1. Sensitivity of the results to the choice of time cut points

Analysis	Alternative	Result
A. Decomposition by cause: split year (2005 → k → 2024)	k = 2012	Share of IMR decline due to perinatal conditions: 64% (2005–2012) vs 30% (2012–2024); perinatal rate change −20% vs −14%
	k = 2013	Share of IMR decline due to perinatal conditions: 56% (2005–2013) vs 34% (2013–2024); perinatal rate change −20% vs −14%
	k = 2014	Share of IMR decline due to perinatal conditions: 57% (2005–2014) vs 29% (2014–2024); perinatal rate change −23% vs −11%
	k = 2015	Share of IMR decline due to perinatal conditions: 56% (2005–2015) vs 10% (2015–2024); perinatal rate change −30% vs −2%
	k = 2016	Share of IMR decline due to perinatal conditions: 59% (2005–2016) vs 6% (2016–2024); perinatal rate change −31% vs −1%
B. All rolling triennia (IMR)	2005–2007	IMR 13.1; Gini 0.095 (0.090–0.102); SII 6.26; RII 1.61; Q4/Q1 1.52
	2006–2008	IMR 12.9; Gini 0.090 (0.084–0.097); SII 5.73; RII 1.56; Q4/Q1 1.49
	2007–2009	IMR 12.6; Gini 0.086 (0.080–0.094); SII 5.03; RII 1.49; Q4/Q1 1.46
	2008–2010	IMR 12.1; Gini 0.083 (0.079–0.093); SII 4.98; RII 1.51; Q4/Q1 1.45
	2009–2011	IMR 11.9; Gini 0.084 (0.079–0.092); SII 4.81; RII 1.50; Q4/Q1 1.44
	2010–2012	IMR 11.5; Gini 0.082 (0.075–0.089); SII 4.37; RII 1.46; Q4/Q1 1.41
	2011–2013	IMR 11.2; Gini 0.075 (0.070–0.084); SII 4.05; RII 1.44; Q4/Q1 1.38
	2012–2014	IMR 10.8; Gini 0.074 (0.068–0.082); SII 4.08; RII 1.46; Q4/Q1 1.36
	2013–2015	IMR 10.3; Gini 0.074 (0.069–0.083); SII 3.97; RII 1.47; Q4/Q1 1.36
	2014–2016	IMR 10.0; Gini 0.076 (0.071–0.086); SII 3.87; RII 1.47; Q4/Q1 1.38
	2015–2017	IMR 9.6; Gini 0.079 (0.072–0.087); SII 3.50; RII 1.44; Q4/Q1 1.36
	2016–2018	IMR 9.3; Gini 0.073 (0.066–0.082); SII 3.18; RII 1.41; Q4/Q1 1.33
	2017–2019	IMR 9.1; Gini 0.078 (0.071–0.088); SII 3.14; RII 1.41; Q4/Q1 1.31
	2018–2020	IMR 8.8; Gini 0.084 (0.077–0.094); SII 3.65; RII 1.51; Q4/Q1 1.38
	2019–2021	IMR 8.6; Gini 0.091 (0.084–0.102); SII 3.82; RII 1.57; Q4/Q1 1.41
	2020–2022	IMR 8.3; Gini 0.092 (0.086–0.103); SII 3.64; RII 1.55; Q4/Q1 1.43
	2021–2023	IMR 8.1; Gini 0.087 (0.081–0.100); SII 3.06; RII 1.46; Q4/Q1 1.37
	2022–2024	IMR 8.3; Gini 0.094 (0.085–0.107); SII 3.44; RII 1.51; Q4/Q1 1.43
C. Beta convergence: split year k	k = 2012	2005–2012: coefficient −4.26 (p 0.020); 2012–2024: 1.10 (p 0.265)
	k = 2013	2005–2013: coefficient −6.05 (p <0.001); 2013–2024: 1.90 (p 0.134)
	k = 2014	2005–2014: coefficient −5.72 (p <0.001); 2014–2024: 1.80 (p 0.238)
	k = 2015	2005–2015: coefficient −5.11 (p <0.001); 2015–2024: 2.39 (p 0.233)
	k = 2016	2005–2016: coefficient −4.15 (p <0.001); 2016–2024: 1.79 (p 0.447)

Analysis	Alternative	Result
D. Kitagawa decomposition by birthweight: alternative periods	2011–2013 vs 2022–2024	IMR 11.23 → 8.30; composition effect +0.58; rate effect −3.51
	2014–2016 vs 2022–2024	IMR 10.00 → 8.30; composition effect +0.53; rate effect −2.22
	2016–2018 vs 2022–2024	IMR 9.31 → 8.30; composition effect +0.51; rate effect −1.52
	2018–2019 vs 2023–2024	IMR 9.00 → 8.24; composition effect +0.33; rate effect −1.09
	2019 vs 2024	IMR 9.19 → 8.50; composition effect +0.29; rate effect −0.97
E. Expected deaths 2020–2024: alternative baselines	2005–2019, Argentina	O/E 1.01; excess +263 (95% PI -626 to +1149)
	2005–2019, Q4 (highest UBN)	O/E 1.12; excess +503 (95% PI +226 to +764)
	2008–2019, Argentina	O/E 1.03; excess +548 (95% PI -298 to +1399)
	2008–2019, Q4 (highest UBN)	O/E 1.13; excess +529 (95% PI +247 to +811)
	2010–2019, Argentina	O/E 1.04; excess +730 (95% PI -258 to +1711)
	2010–2019, Q4 (highest UBN)	O/E 1.14; excess +557 (95% PI +205 to +886)
	2012–2019, Argentina	O/E 1.03; excess +653 (95% PI -627 to +1900)
	2012–2019, Q4 (highest UBN)	O/E 1.12; excess +487 (95% PI +26 to +917)
	2005–2018, Argentina	O/E 1.02; excess +463 (95% PI -484 to +1362)
	2005–2018, Q4 (highest UBN)	O/E 1.14; excess +569 (95% PI +291 to +835)
F. Change in IMR slope: alternative hinge year	hinge at 2017	APC before −3.1; APC after −2.4; p for change 0.179
	hinge at 2018	APC before −3.1; APC after −2.1; p for change 0.102
	hinge at 2019	APC before −3.1; APC after −1.8; p for change 0.087
	hinge at 2020	APC before −3.1; APC after −1.1; p for change 0.034

Note: Main analyses used: split year 2014 (A, C); triennia 2005–2007, 2013–2015, 2017–2019 and 2022–2024 (B, D); baseline 2005–2019 (E); hinge at 2019 (F). Gini: bootstrap 95% CI (200 replicates). SII: slope index of inequality (deaths per 1 000 live births); RII: relative index of inequality; Q4/Q1: ratio of IMR between extreme UBN quartiles; O/E: observed/expected deaths; PI: prediction interval; APC: annual percent change.

Table S2. Robustness of the change in trend

Analysis	Series	Estimate
A. Date of breakpoint (Bai–Perron, log rate), 95% CI	IMR 1980–2024	Last year of previous segment 1996 (1994–1997)
	IMR 1980–2024	Last year of previous segment 2003 (2002–2005)
	IMR 1980–2024	Last year of previous segment 2019 (2015–2020)
	IMR 2005–2024	Last year of previous segment 2020 (2018–2021)
	NMR 2005–2024	Last year of previous segment 2014 (2007–2015)
	PNMR 2005–2024	Last year of previous segment 2019 (2017–2020)
B. Segmented quasi-Poisson regression, 2005–2024	IMR	Breakpoint 2022.8 (2021.7–2023.9); APC before −3.1 (−3.3 to −2.8); APC after 6.2 (−4.1 to 17.7); Davies test p <0.001
	NMR	Breakpoint 2022.0 (2020.3–2023.6); APC before −2.6 (−2.9 to −2.4); APC after 1.6 (−3.0 to 6.5); Davies test p <0.001
	PNMR	Breakpoint 2007.0 (2002.1–2011.9); APC before 0.8 (−12.2 to 15.7); APC after −4.1 (−4.6 to −3.5); Davies test p <0.001
C. Change in slope after 2019 (hinge quasi-Poisson model)	IMR, 2005–2024	APC before −3.1; APC after −1.8; p for change 0.087
	IMR, excluding 2020–2021	APC before −3.0; APC after −1.6; p for change 0.055
	NMR, 2005–2024	APC before −2.7; APC after −1.4; p for change 0.066
	NMR, excluding 2020–2021	APC before −2.7; APC after −1.3; p for change 0.048
	PNMR, 2005–2024	APC before −4.0; APC after −2.7; p for change 0.442
	PNMR, excluding 2020–2021	APC before −3.7; APC after −2.4; p for change 0.306

Note: APC: annual percent change. Breakpoints in A are reported as the last year before the change. B: estimates in the postneonatal series are unstable because of the 2020–2021 drop. C: models with log live births as offset.

Table S3. Observed and expected infant deaths, 2020–2024, according to the 2005–2019 trend

Group	Indicator	Baseline	Year	Observed	Expected (95% PI)	Excess	O/E
Argentina	IMR	2005–2019	2020	4505	4604 (4282–4930)	-99	0.98
Argentina	IMR	2005–2019	2021	4238	4435 (4121–4751)	-197	0.96
Argentina	IMR	2005–2019	2022	4162	4021 (3724–4323)	+141	1.04
Argentina	IMR	2005–2019	2023	3689	3629 (3346–3922)	+60	1.02
Argentina	IMR	2005–2019	2024	3513	3154 (2889–3432)	+359	1.11
Argentina	IMR	2005–2019	2020–2024	20107	19844 (18958–20733)	+263 (-626 to +1149)	1.01
Argentina	NMR	2005–2019	2020	3303	3163 (2975–3354)	+140	1.04
Argentina	NMR	2005–2019	2021	3029	3058 (2872–3252)	-29	0.99
Argentina	NMR	2005–2019	2022	2864	2782 (2604–2964)	+82	1.03
Argentina	NMR	2005–2019	2023	2525	2519 (2349–2700)	+6	1.00
Argentina	NMR	2005–2019	2024	2476	2197 (2036–2365)	+279	1.13
Argentina	NMR	2005–2019	2020–2024	14197	13720 (13189–14271)	+477 (-74 to +1008)	1.03
Argentina	PNMR	2005–2019	2020	1202	1443 (1248–1645)	-241	0.83
Argentina	PNMR	2005–2019	2021	1209	1381 (1190–1585)	-172	0.88
Argentina	PNMR	2005–2019	2022	1298	1243 (1059–1438)	+55	1.04
Argentina	PNMR	2005–2019	2023	1164	1114 (941–1299)	+50	1.04
Argentina	PNMR	2005–2019	2024	1036	962 (800–1134)	+74	1.08
Argentina	PNMR	2005–2019	2020–2024	5909	6145 (5610–6710)	-236 (-801 to +299)	0.96
Argentina	IMR	2010–2019	2020	4505	4523 (4220–4836)	-18	1.00
Argentina	IMR	2010–2019	2021	4238	4343 (4034–4654)	-105	0.98
Argentina	IMR	2010–2019	2022	4162	3924 (3621–4233)	+238	1.06
Argentina	IMR	2010–2019	2023	3689	3529 (3247–3830)	+160	1.05
Argentina	IMR	2010–2019	2024	3513	3057 (2788–3343)	+456	1.15
Argentina	IMR	2010–2019	2020–2024	20107	19377 (18396–20365)	+730 (-258 to +1711)	1.04
Q1 (lowest UBN)	IMR	2005–2019	2020	1027	1084 (983–1189)	-57	0.95
Q1 (lowest UBN)	IMR	2005–2019	2021	956	1055 (952–1161)	-99	0.91
Q1 (lowest UBN)	IMR	2005–2019	2022	1006	974 (873–1078)	+32	1.03
Q1 (lowest UBN)	IMR	2005–2019	2023	920	877 (781–978)	+43	1.05
Q1 (lowest UBN)	IMR	2005–2019	2024	792	766 (679–859)	+26	1.03
Q1 (lowest UBN)	IMR	2005–2019	2020–2024	4701	4756 (4464–5060)	-55 (-359 to +237)	0.99
Q2	IMR	2005–2019	2020	1910	2052 (1882–2229)	-142	0.93
Q2	IMR	2005–2019	2021	1885	1964 (1791–2141)	-79	0.96
Q2	IMR	2005–2019	2022	1731	1759 (1597–1929)	-28	0.98
Q2	IMR	2005–2019	2023	1523	1587 (1434–1741)	-64	0.96
Q2	IMR	2005–2019	2024	1506	1400 (1254–1552)	+106	1.08

Group	Indicator	Baseline	Year	Observed	Expected (95% PI)	Excess	O/E
Q2	IMR	2005–2019	2020–2024	8555	8761 (8279–9246)	-206 (-691 to +276)	0.98
Q3	IMR	2005–2019	2020	501	519 (452–589)	-18	0.97
Q3	IMR	2005–2019	2021	477	501 (434–570)	-24	0.95
Q3	IMR	2005–2019	2022	455	458 (395–526)	-3	0.99
Q3	IMR	2005–2019	2023	396	409 (349–472)	-13	0.97
Q3	IMR	2005–2019	2024	367	352 (297–411)	+15	1.04
Q3	IMR	2005–2019	2020–2024	2196	2239 (2053–2434)	-43 (-238 to +143)	0.98
Q4 (highest UBN)	IMR	2005–2019	2020	1041	945 (852–1043)	+96	1.10
Q4 (highest UBN)	IMR	2005–2019	2021	899	916 (822–1016)	-17	0.98
Q4 (highest UBN)	IMR	2005–2019	2022	954	820 (730–914)	+134	1.16
Q4 (highest UBN)	IMR	2005–2019	2023	839	755 (668–847)	+84	1.11
Q4 (highest UBN)	IMR	2005–2019	2024	840	633 (556–714)	+207	1.33
Q4 (highest UBN)	IMR	2005–2019	2020–2024	4573	4070 (3809–4347)	+503 (+226 to +764)	1.12
Q4 (highest UBN)	IMR (all causes)	2005–2019	2020–2024	4573	4071 (3809–4348)	+502 (+225 to +764)	1.12
Q4 (highest UBN)	NMR	2005–2019	2020–2024	3126	2769 (2580–2965)	+357 (+161 to +546)	1.13
Q4 (highest UBN)	PNMR	2005–2019	2020–2024	1447	1307 (1167–1456)	+140 (-9 to +280)	1.11
Q4 (highest UBN)	Perinatal conditions (P)	2005–2019	2020–2024	2429	2059 (1933–2194)	+370 (+235 to +496)	1.18
Q4 (highest UBN)	Congenital anomalies (Q)	2005–2019	2020–2024	1224	1208 (1061–1363)	+16 (-139 to +163)	1.01
Q4 (highest UBN)	Respiratory (J)	2005–2019	2020–2024	235	303 (195–441)	-68 (-206 to +40)	0.78
Q4 (highest UBN)	Ill-defined (R)	2005–2019	2020–2024	186	185 (143–232)	+1 (-46 to +43)	1.01

Note: Expected deaths from quasi-Poisson regression (log live births offset, linear year) fitted to the baseline period; 95% prediction intervals (PI) from 10,000 simulations including parameter uncertainty and overdispersion. Excess for 2020–2024 shown with 95% PI. O/E: observed/expected.

Table S4. Infant, neonatal and postneonatal mortality and inequality measures by triennium, Argentina, 2005–2024

Indicator	Triennium	Rate	Q1	Q4	Q4/Q1	Q4−Q1	Gini (95% CI)	CIx (95% CI)	SII (95% CI)	RII (95% CI)
IMR	2005–2007	13.13	11.00	16.74	1.52	5.74	0.095 (0.090–0.102)	−0.075 (−0.082 to −0.069)	6.26 (3.35 to 9.18)	1.61 (1.32–1.97)
IMR	2013–2015	10.34	8.92	12.10	1.36	3.18	0.074 (0.068–0.083)	−0.060 (−0.067 to −0.053)	3.97 (2.42 to 5.52)	1.47 (1.28–1.68)
IMR	2017–2019	9.12	7.97	10.40	1.31	2.44	0.078 (0.072–0.087)	−0.054 (−0.062 to −0.047)	3.14 (1.45 to 4.83)	1.41 (1.19–1.67)
IMR	2022–2024	8.30	7.24	10.37	1.43	3.13	0.094 (0.086–0.106)	−0.064 (−0.073 to −0.053)	3.44 (1.59 to 5.28)	1.51 (1.23–1.86)
NMR	2005–2007	8.58	7.41	10.80	1.46	3.39	0.100 (0.093–0.110)	−0.074 (−0.081 to −0.066)	4.02 (1.76 to 6.27)	1.60 (1.26–2.02)
NMR	2013–2015	7.03	6.24	8.09	1.30	1.85	0.075 (0.069–0.086)	−0.058 (−0.067 to −0.049)	2.58 (1.09 to 4.06)	1.44 (1.19–1.74)
NMR	2017–2019	6.24	5.52	7.01	1.27	1.49	0.086 (0.079–0.098)	−0.053 (−0.063 to −0.044)	2.11 (0.60 to 3.62)	1.40 (1.13–1.75)
NMR	2022–2024	5.75	5.14	6.83	1.33	1.69	0.097 (0.087–0.113)	−0.053 (−0.067 to −0.040)	2.01 (0.42 to 3.61)	1.42 (1.10–1.84)
PNMR	2005–2007	4.55	3.59	5.94	1.65	2.35	0.115 (0.106–0.129)	−0.078 (−0.090 to −0.067)	2.25 (1.10 to 3.40)	1.64 (1.30–2.06)
PNMR	2013–2015	3.31	2.68	4.01	1.49	1.32	0.092 (0.083–0.108)	−0.066 (−0.078 to −0.055)	1.39 (0.73 to 2.04)	1.52 (1.27–1.83)
PNMR	2017–2019	2.87	2.44	3.39	1.39	0.95	0.093 (0.084–0.112)	−0.057 (−0.071 to −0.043)	1.03 (0.32 to 1.74)	1.43 (1.14–1.80)
PNMR	2022–2024	2.55	2.09	3.54	1.69	1.44	0.118 (0.108–0.143)	−0.086 (−0.106 to −0.067)	1.42 (0.77 to 2.07)	1.75 (1.39–2.21)

Note: Rates per 1,000 live births. Q1–Q4: quartiles of provinces by percentage of persons with unsatisfied basic needs (UBN), 2010 census (6 provinces each). Gini: between-province Gini index (bootstrap 95% CI, 1,000 replicates). CIx: concentration index, provinces ranked from highest to lowest UBN (negative values indicate concentration of deaths in provinces with worse living conditions). SII: slope index of inequality (deaths per 1,000 live births); RII: relative index of inequality; both from regression on the births-weighted ridit of UBN.

Table S5. Infant mortality by province, 2005–2007 and 2022–2024, and beta convergence

Province	UBN quartile	% UBN 2010	% UBN 2022	IMR 2005–2007	IMR 2022–2024	APC 2005–2014	APC 2014–2024	p 2014–2024
Corrientes	Q4	19.8	15.2	17.1	13.0	−1.8	−1.7	0.111
Formosa	Q4	25.3	15.1	23.3	12.1	−5.5	−2.3	0.161
La Rioja	Q3	15.5	10.6	13.6	10.8	−0.6	−0.7	0.571
Chaco	Q4	23.3	14.3	20.0	10.7	−6.6	−1.5	0.169
Tucumán	Q3	16.6	12.0	14.2	10.0	−0.9	−2.7	0.010
Entre Ríos	Q2	11.6	8.5	12.6	9.8	−3.5	−0.7	0.476
Santa Cruz	Q1	9.8	8.0	13.1	9.6	−3.9	−1.3	0.451
Salta	Q4	23.8	16.9	14.8	9.5	−1.5	−2.7	0.015
Santiago del Estero	Q4	22.8	12.8	12.5	9.3	−0.8	−1.0	0.588
Misiones	Q4	19.3	11.1	15.4	8.8	−5.0	−0.6	0.535
San Juan	Q3	14.1	11.2	14.6	8.7	−4.2	−1.6	0.151
Santa Fe	Q1	9.5	7.4	11.7	8.3	−1.9	−1.2	0.079
Buenos Aires	Q2	11.3	9.0	13.1	7.9	−2.4	−2.9	<0.001
Río Negro	Q2	11.8	8.2	9.5	7.9	0.0	−0.3	0.669
La Pampa	Q1	5.8	4.2	11.1	7.5	−4.7	−1.7	0.411
Catamarca	Q3	14.6	8.8	15.7	7.4	−6.7	−3.3	0.067
Córdoba	Q1	8.8	6.6	12.1	7.3	−3.3	−2.8	0.003
Jujuy	Q3	18.2	13.2	16.1	7.1	−4.6	−4.9	<0.001
San Luis	Q2	10.8	9.1	14.8	7.0	−5.0	−4.2	0.004
Mendoza	Q1	10.4	9.2	11.5	6.7	−2.7	−3.3	0.003
Chubut	Q2	10.8	7.0	11.4	6.4	−3.2	−2.8	0.149
Neuquén	Q2	12.5	8.8	9.8	6.1	−1.0	−4.3	0.053
Tierra del Fuego	Q3	14.6	15.5	9.3	5.9	−0.8	−3.2	0.122
CABA	Q1	7.0	6.0	8.2	5.5	0.5	−4.3	0.006

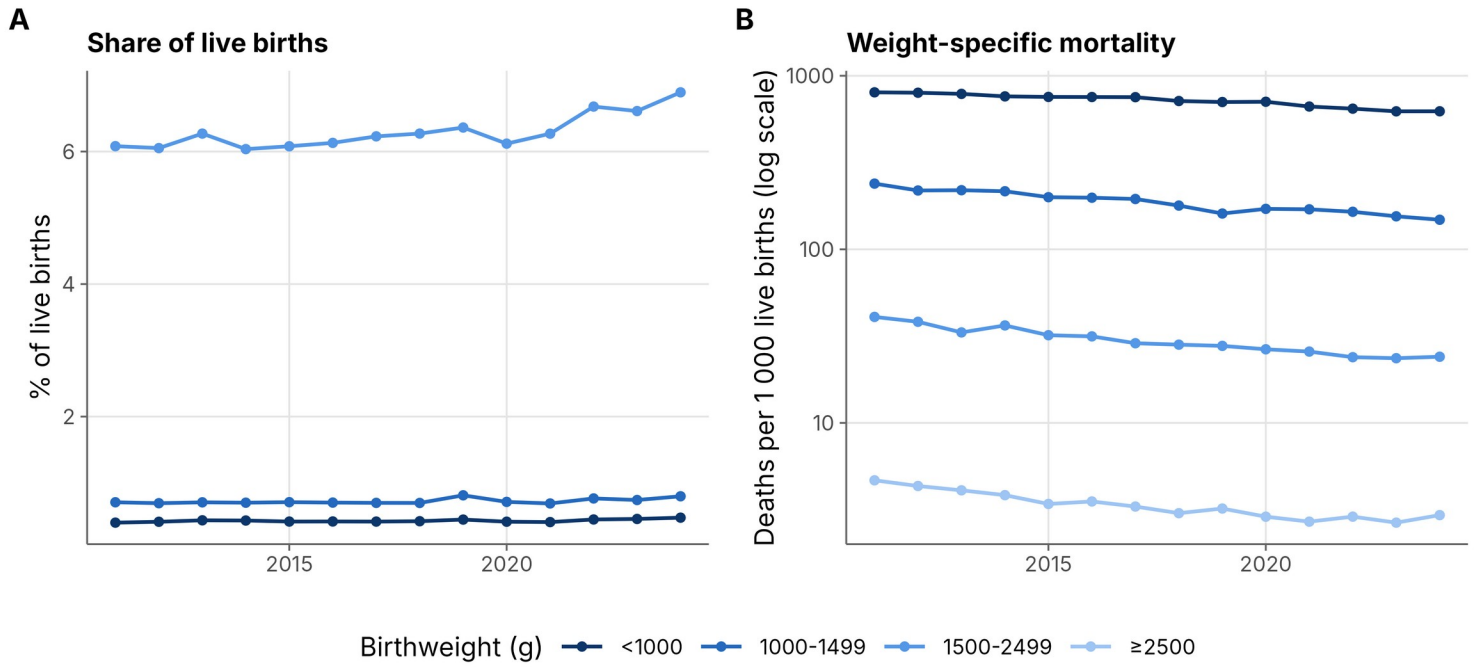
Note: Beta convergence (regression of provincial APC on log initial IMR): 2005–2014 coefficient −5.72 (95% CI −8.55 to −2.88; R^2 0.44); 2014–2024 coefficient 1.80 (−1.28 to 4.88; R^2 0.06). UBN: percentage of persons with unsatisfied basic needs (INDEC).

Table S6. Infant mortality rate and annual rate of reduction (ARR) in countries of the Americas, UN IGME estimates

Country (ISO3)	IMR 2015	IMR 2019	IMR 2024	ARR 2010–2015	ARR 2015–2019	ARR 2019–2024
CUB	4.8	5.4	6.8	0.1	−2.9	−4.5
CRI	7.8	7.9	9.1	3.0	−0.4	−2.8
CAN	4.7	4.5	4.7	1.3	0.8	−0.9
USA	5.8	5.5	5.5	1.6	1.1	0.1
URY	7.5	6.2	6.2	4.1	4.6	0.3
CHL	6.8	6.0	5.8	1.9	3.0	0.8
PAN	13.6	12.6	12.0	1.8	2.0	1.0
JAM	17.2	17.2	16.4	0.3	−0.1	1.0
BLZ	13.1	11.4	10.8	3.4	3.5	1.0
BRA	13.7	12.9	12.3	3.0	1.5	1.1
ECU	12.9	11.7	10.9	4.3	2.5	1.3
ARG	10.0	8.9	8.0	4.2	3.0	2.1
DOM	31.0	30.9	27.7	−0.3	0.1	2.1
TTO	20.6	19.2	17.0	1.8	1.8	2.4
MEX	15.1	13.7	12.1	2.4	2.3	2.6
PER	13.8	12.5	11.0	3.6	2.4	2.6
HTI	49.7	44.7	39.0	9.3	2.6	2.7
SUR	19.0	17.1	14.8	2.2	2.6	3.0
BRB	11.7	10.6	9.1	1.6	2.6	3.1
GUY	30.4	27.1	23.2	2.2	2.9	3.1
COL	13.9	12.2	10.5	3.1	3.1	3.1
HND	17.0	14.9	12.8	3.3	3.3	3.2
PRY	19.3	17.1	14.5	3.1	3.1	3.2
GTM	23.1	20.2	17.2	3.9	3.4	3.2
SLV	12.6	10.6	8.8	4.4	4.1	3.7
BOL	23.5	18.4	14.2	5.9	6.0	5.2
BHS	12.3	16.0	11.1	1.8	−6.6	7.3
ARG (DEIS)	9.7	9.2	8.5	4.1	1.3	1.5

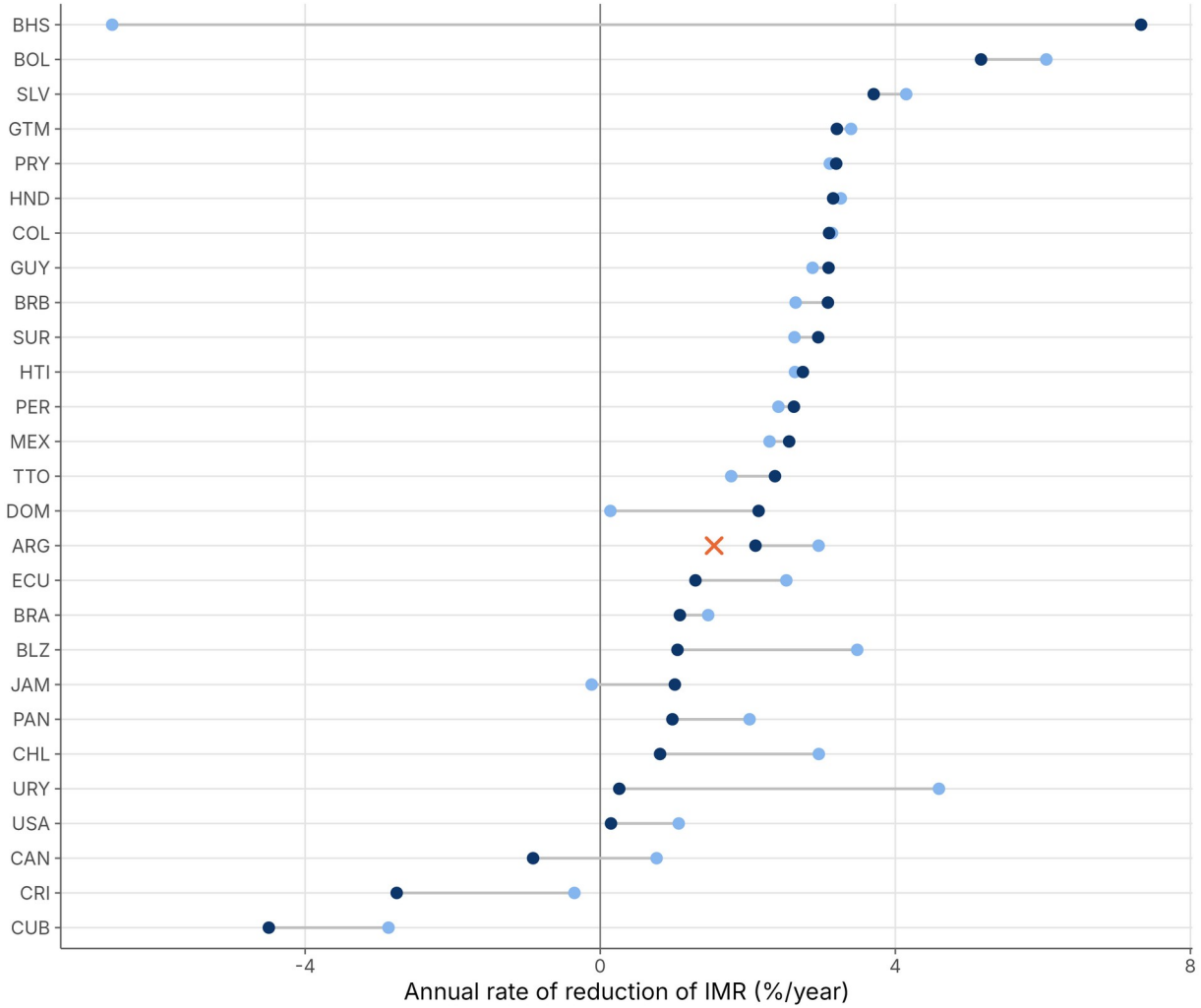
Note: UN IGME model-based (smoothed) estimates accessed through the UNICEF data warehouse (CME dataflow) on 6 October 2026. ARR = −ln(IMR_t2/IMR_t1)/(t2−t1) × 100; negative values indicate increasing mortality. Last row: Argentina, official vital statistics (DEIS). Nicaragua and Venezuela were not included.

Figure S2. Birthweight distribution of live births (A) and birthweight-specific infant mortality (B), Argentina, 2011–2024



Note: Deaths and births with unspecified birthweight distributed proportionally within each year. Source: DEIS vital statistics yearbooks.

Figure S3. Annual rate of reduction of the infant mortality rate in countries of the Americas, 2015–2019 (light points) and 2019–2024 (dark points), UN IGME estimates



Note: Negative values indicate increasing mortality. ×: Argentina, official vital statistics (DEIS), 2019–2024. Source: UN IGME, accessed 6 October 2026.